TU N TRAN

Houston, TX | (346) 515-0455 | trantunhi99@gmail.com | linkedin.com/in/tuntran271299 |
Portfolio (AI + Data Science): https://alexthefirst99.github.io/web/

SUMMARY

Graduate student in Engineering Data Science with experience applying statistical modeling, machine learning, and data visualization to biological and engineering problems. Skilled in Python, R, SQL (in progress), and cloud-based workflows. Experienced in building end-to-end pipelines, cleaning and structuring large datasets, and developing interactive dashboards. Seeking a Data Science internship to leverage analytical skills and programming expertise for data-driven decision making.

EDUCATION

University of Houston, Cullen College of Engineering – Houston, TX (Expected December 2026)

Master of Science in Engineering Data Science and AI GPA: 3.63

University of Massachusetts Lowell – Lowell, MA Aug 2017 – Aug 2021

Bachelor of Science in Mathematics, Minor in Finance GPA: 3.135

Dean's List 2020–2021

RELEVANT PROJECTS

Loki Interactive Inference Platform – Houston Methodist Spring 2026

- Developed a full-stack web platform on top of Loki, an AI framework for spatial transcriptomics and histopathology analysis.
- Built a Dash/React-based interface for launching deep learning inference workflows on gigapixel whole-slide H&E images from the browser.
- Connected frontend interactions with backend pipelines for nuclei segmentation and cell-type annotation, removing the need for local setup or CLI-based execution.
- Implemented interactive visualization of whole-slide images with scalable rendering and color-coded biological overlays.
- Added an LLM-assisted interpretation feature to help users query segmented cell populations and tissue microenvironments in natural language.

Visualization Website for Gene Expression (Mjolnir) – Houston Methodist Summer 2025

- Built an interactive data visualization platform using Python (Dash, Plotly) for exploration of multimodal datasets
- Integrated gigapixel histology images with gene expression and pathway analysis to enable intuitive exploration by non-technical users
- Contributed to a Nature Communications publication on multimodal spatial data analysis

Visual–Omics Model Integration & Data Curation – Houston Methodist Summer 2025

- Curated and structured multimodal datasets linking histology images with spatial gene expression across 32 organs for machine learning workflows
- Built Python scripts to evaluate model predictions and quantify alignment with known biological signals
- Performed data validation and consistency checks to improve reliability of downstream model training and benchmarking
- Contributed to a Nature Methods publication on multimodal foundation models

Macrophage Analysis – Houston Methodist Spring 2023

- Applied unsupervised clustering (Leiden) and UMAP to identify and interpret immune cell subtypes from single-cell datasets
- Quantified treatment effects by analyzing differential marker gene expression across macrophage populations
- Developed Python pipelines for data preprocessing, normalization, and reproducible exploratory analysis

WORK EXPERIENCE

Research Assistant – Cardiovascular Regeneration Program

Houston Methodist, Dr. Guangyu Wang Lab — Dec 2022 – Present

- Built data pipelines for large-scale single-cell and spatial omics datasets to support machine learning model development and evaluation
- Applied unsupervised learning and statistical analysis to identify patterns in high-dimensional biological data
- Assessed model predictions and validated outputs against biological signals to ensure reliability and interpretability
- Developed interactive dashboards for visualization and interpretation of model results by research teams
- Contributed to publications in Nature Methods (multimodal foundation models) and Nature Communications (cell-level integration of spatial transcriptomics and histology)

Marketing & Data Intern

Viet Power Solutions JSC, Vietnam — Jan 2022 – Aug 2022

- Monitored digital engagement metrics with Google Analytics and optimized campaigns for +39.5% performance improvement.
- Designed and launched targeted LinkedIn, Facebook, and Google Ads campaigns, raising user engagement by 50%.
- Produced clear, data-driven reports to inform leadership and refine product strategy.

PUBLICATIONS

- Chen W., Zhang P., Tran T. N., et al. (2025). A visual–omics foundation model to bridge histopathology with spatial transcriptomics. Nature Methods. doi:10.1038/s41592-025-02707-1
- Zhang P., Chen W., Tran T. N., et al. (2025). Thor: a platform for cell-level investigation of spatial transcriptomics and histology. Nature Communications, 16, 7178. doi:10.1038/s41467-025-62593-1

SKILLS

- Programming: Python (Pandas, NumPy, SciPy, scikit-learn, Plotly, Dash), R (tidyverse, Seurat), HTML/CSS
- Data Science: Data wrangling, statistical modeling, clustering, dimensionality reduction, hypothesis testing, pathway enrichment
- Machine Learning: scikit-learn, trajectory inference, model diagnostics, evaluation metrics
- Data Engineering: Snakemake, Git, Linux, AWS (EC2/S3), data preprocessing and automation
- Visualization: Dash/Plotly, Matplotlib, Seaborn, interactive dashboards and reports
- In Progress: SQL, A/B testing, Tableau/Looker